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Reinforcement Learning with Continuous Actions Under Unmeasured Confounding

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ABSTRACT
This article addresses the challenge of offline policy learning in continuous action spaces when unmeasured confounders are present. While most existing research focuses on policy evaluation within partially observable Markov decision processes (POMDPs) and assumes discrete action spaces, we advance this field by establishing a novel identification result to enable the nonparametric estimation of policy value for a given target policy under an infinite-horizon framework. Leveraging this identification, we develop a minimax estimator and introduce a policy-gradient-based algorithm to identify the in-class optimal policy that maximizes the estimated policy value. Furthermore, we provide theoretical results regarding the consistency, finite-sample error bound, and regret bound of the resulting optimal policy. Extensive simulations and a real-world application using the German Family Panel data demonstrate the effectiveness of our proposed methodology. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

In practical applications of reinforcement learning (RL) (Sutton and Barto 2018), the evaluation and optimization of policies using only pre-collected datasets has become essential. This need arises from the potential costs and safety risks associated with frequent interactions with the environment, such as real-world testing in autonomous driving (Zhu et al. 2020) and treatment selection in precision medicine (Luckett et al. 2019; Zhou, Li, and Zhu 2024). As a result, there has been a growing interest in offline RL (Precup 2000; Levine et al. 2020), which focuses on policy evaluation and optimization without requiring additional environment interactions.

Recent years have seen significant advancements in both off-policy evaluation (OPE) and off-policy learning (OPL). In OPE, popular methodologies include value-based approaches (Le, Voloshin, and Yue 2019; Liao, Klasnja, and Murphy 2021), importance sampling techniques (Liu et al. 2018; Xie, Ma, and Wang 2019), and doubly robust methods that integrate these two estimators (Uehara, Huang, and Jiang 2020; Kallus et al. 2022). Additionally, confidence intervals for a target policy have been developed using bootstrapping (Hao et al. 2021), asymptotic properties (Shi et al. 2022), and finite-sample error bounds (Zhou et al. 2023). In OPL, various algorithms have been proposed, demonstrating notable success across both discrete (Mnih et al. 2015; Luckett et al. 2019; Zhou, Zhu, and Qu 2024) and continuous action spaces (Kumar et al. 2020; Cai et al. 2021; Li, Zhou, and Zhu 2023; Zhou 2024).

A common assumption in the works mentioned above is the absence of unmeasured confounders. Specifically, these studies assume that all state variables are fully observed, with no unmeasured variables that might confound the observed actions. However, this assumption is unverifiable with offline data and is often violated in practical settings. Real-world instances include the influence of genetic factors in personalized medicine (Fröhlich et al. 2018) and the complexities of path planning in robotics (Zhang, Kumor, and Bareinboim 2020). To address these challenges, a recent line of research has focused on OPE within the framework of a confounded partially observable Markov Decision Process (POMDP), where the behavior policy generating the batch data may depend on unobserved state variables (Zhang and Bareinboim 2016).

Various methods have been developed to identify the true policy value with the presence of unmeasured confounders. Within the contextual bandit setting (i.e., one decision point), causal inference frameworks have been established to identify the treatment effect by sensitivity analysis (Bonvini and Kennedy 2022), and through the use of instrumental variables (Cui and Tchetgen Tchetgen 2021) and negative controls (Miao, Geng, and Tchetgen Tchetgen 2018; Cui et al. 2023). Existing methods for extensions to multiple decision points or the infinite horizon setting fall into two main categories. The first type considers iid confounders in the dynamic system and thus preserves the Markovian property (Zhang and Bareinboim 2016). Under this framework, OPE methods have been formulated under various identification conditions, such as partial identification via sensitivity analysis (Kallus and Zhou 2020) and approaches that use instrumental variables or mediators (Li, Luo, and Zhang 2021; Shi et al. 2022). The second category explores the estimation of policy values in more general confounded POMDP models,

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where the Markovian assumption does not hold. These methods span a wide array of strategies, including the use of proxy variables (Shi et al. 2022; Miao, Qi, and Zhang 2022; Uehara et al. 2024) or instrumental variables (Fu et al. 2022; Xu et al. 2023), spectral methods for undercomplete POMDPs (Jin et al. 2020), and techniques focusing on predictive state representation (Cai, Yang, and Wang 2022; Guo et al. 2022).

However, there are two less investigated issues in confounded POMDP settings. First, existing methods mainly focus on discrete action spaces (Miao, Qi, and Zhang 2022; Shi et al. 2022; Bennett and Kallus 2023) despite many real-world scenarios requiring decision making on a continuous action space (Lillicrap et al. 2015). A straightforward workaround in adapting existing methods to continuous domains is to discretize the continuous action space. However, this approach either introduces significant bias when using coarse discretization (Cai et al. 2021) or encounters the curse of dimensionality when applied to fine grids (Chou, Maturana, and Scherer 2017). Second, the challenge of learning an optimal policy from a batch dataset, particularly in the presence of unmeasured confounders, is paramount in various fields, including personalized medicine (Lu et al. 2022) and robotics (Brunke et al. 2022). A majority of existing methods, however, focus on policy evaluation rather than optimization. There are some recent efforts to address this gap. For instance, Qi, Miao, and Zhang (2023) explore policy learning under the contextual bandit setting, and Hong, Qi, and Xu (2023) investigate policy gradient methods under finite-horizon confounded POMDPs. In terms of the infinite-horizon, Kallus and Zhou (2021) and Fu et al. (2022) consider a restrictive memoryless confounders setting where the Markovian property is preserved, while Guo et al. (2022) and Lu et al. (2022) focus on theoretical properties of the induced estimator under more general confounded POMDP settings, yet practical computational algorithms remain elusive. Thus, the development of computationally viable policy learning algorithms under infinite-horizon confounded MDPs, where the Markovian property is violated, remains a significant challenge.

Motivated by these, in this article, we study the policy learning with continuous actions for confounded POMDPs over an infinite horizon. Our main contribution to the literature is threefold. First, relying on some time-dependent proxy variables, we extend the proximal causal inference framework (Miao, Geng, and Tchetgen Tchetgen 2018; Tchetgen Tchetgen et al. 2020) to infinite horizon setting, and establish a nonparametric identification result for OPE using V-bridge functions with continuous actions for confounded POMDPs. Leveraging the identification result, we propose an unbiased minimax estimator for the V-bridge function and introduce a computationally efficient Fitted-Q Evaluation (FQE)-type algorithm for estimating the policy value. Second, we develop a novel policy gradient algorithm that searches the optimal policy within a specified policy class by maximizing the estimated policy value. The proposed algorithm is tailored for continuous action spaces and provides enhanced interpretability of the optimal policy. Third, we thoroughly investigate the theoretical properties of the proposed methods on both policy evaluation and policy learning, including the estimator consistency, finite-sample bound for performance error, and sub-optimality of the induced optimal policy. We validate our proposed method through extensive numerical experiments and apply it to the German Family Panel (Pairfam) dataset (Brüderl et al. 2023), where we aim at identifying optimal strategies to enhance long-term relationship satisfaction.

2. Preliminaries

We consider an infinite-horizon confounded Partially Observable Markov Decision Process (POMDP) defined as $\mathcal{M} = \{\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathbf{P}, R, \gamma\}$, where $\mathcal{S}$ and $\mathcal{O}$ denote the unobserved and observed continuous state space, respectively, $\mathcal{A}$ is the action space, $\mathbf{P} : \mathcal{S} \times \mathcal{O} \times \mathcal{A} \to \Delta(\mathcal{S} \times \mathcal{O})$ is the unknown transitional kernel, $R : \mathcal{S} \times \mathcal{O} \times \mathcal{A} \to \mathbb{R}$ is a bounded reward function, and $\gamma \in [0, 1)$ is the discounted factor that balances the immediate and future rewards. $\mathcal{O}$ can also be treated as the observation space in the classical POMDP, then the process $\mathcal{M}$ can be summarized as $\{S_t, O_t, A_t, R_t\}_{t=1}^T$, with S_t and O_t as unobserved and observed state variables, A_t as the action, and R_t as the reward.

The objective of policy learning is to search an optimal policy, π^*, which maximizes the expected discounted sum of rewards, using batch data obtained from a behavior policy π^b. We assume the batch data consists of n iid trajectories, that is,

$$\mathcal{D}_n = \{\mathcal{D}^i\}_{i=1}^n = \{O_0^i, A_0^i, R_0^i, O_1^i, \ldots, O_T^i, A_T^i, R_T^i, O_{T+1}^i\}_{i=1}^n,$$

where the length of trajectory T is assumed to be fixed for simplicity, and the information on unobserved state S_t^i is not available. In this article, we focus on scenarios where the behavior policy, π^b, maps both unobserved and observed state spaces to the action space, that is, $\pi^b : \mathcal{S} \times \mathcal{O} \to \mathcal{A}$. Meanwhile, the target optimal policy only depends on the observed state: $\pi^* : \mathcal{O} \to \mathcal{A}$. For a given target policy $\pi : \mathcal{O} \to \mathcal{A}$, its state-value function is denoted as

$$V^\pi(s, o) = \mathbb{E}_\pi\Big[\sum_{k=0}^{\infty} \gamma^k R_{t+k} \mid S_t = s, O_t = o\Big], \tag{2.1}$$

where $\mathbb{E}_\pi$ denotes the expectation with respect to the distribution whose actions at each decision point t follow policy π. Our goal is to use the batch data $\mathcal{D}_n$ to find the optimal policy π^* which maximizes the target policy value defined as

$$J(\pi^*) = \arg\max_\pi \mathbb{E}[V^\pi(S_0, O_0)], \tag{2.2}$$

with $\mathbb{E}$ representing the expectation in accordance with the behavior policy. Due to the unobserved state S_t, standard policy learning approaches based on the Bellman optimality equation yield biased estimates. Thus, we first introduce an identification result to estimate the policy value for any target policy π with the help of some proxy variables. Subsequently, we employ policy gradient techniques to find the optimal policy π^*.

3. Methodology

Inspired by the proximal causal inference framework introduced by Tchetgen Tchetgen et al. (2020), we initially present the identification result for policy evaluation in Section 3.1. Following this, its associated minimax estimator for any given target policy π is discussed in Section 3.2. Building on these policy evaluation findings, we detail a policy-gradient based approach in Section 3.3 to identify the optimal policy within a policy class by maximizing the estimated policy value.

3.1. Identification Results

In this section, we present the identification result for confounded POMDP setting. The derived identification equation serves as a foundation for off-policy evaluation (OPE) in the presence of unmeasured confounding, while also ensures the existence of bridge functions.

Building on the proximal causal inference framework proposed by Tchetgen Tchetgen et al. (2020), we further assume the observation of reward-inducing proxy variables, W_t, that relate to the action A_t solely through $\{S_t, O_t\}$. In practical scenarios, W_t could represent environmental factors correlated with the outcome R_t, but remain unaffected by A_t. For example, in healthcare applications, W_t might include the choice of doctors or hospitals administering the treatment. Alternatively, it could consist of variables that are either not accessible for decision-making due to privacy concerns or that become available only after the treatment. As for family panel studies (Brüderl et al. 2023), W_t can be selected as the variables related to housing conditions, working environments and educational backgrounds of family members, while the action can be defined as the time spent with family. A representative Directed Acyclic Graph (DAG) of this can be seen in the left panel Figure 1. The observed data for the confounded POMDP then have the form of

$$\mathcal{D}_n = \{\mathcal{D}^i\}_{i=1}^n = \{O_0^i, W_0^i, A_0^i, R_0^i, O_1^i, \ldots, O_T^i, W_T^i, A_T^i, R_T^i, O_{T+1}^i, W_{T+1}^i\}_{i=1}^n.$$

The left panel of Figure 1 can be viewed as a specific example of the proximal causal inference framework proposed by Tchetgen Tchetgen et al. (2020) and Cui et al. (2023). In our representation, we consider the previously observed state-action pair (O_{t-1}, A_{t-1}) as the action-inducing proxy Z_t. Therefore, all arrows related to Z_t from the original framework can be removed.

In contrast to the classical POMDP setting discussed in (Shi et al. 2022; Bennett and Kallus 2023) and illustrated in the right panel of Figure 1, where immediate reward and the behavior policy depends solely on unobserved states, our proposed causal framework allows the immediate reward and behavior policy to be influenced by both observed and unobserved states with the help of additional reward-proxy variables W_t. Such modification more accurately captures real-world scenarios in which observable state variables can significantly affect decision-making during batch data collection process. A notable example is in precision medicine, where observable variables, such as laboratory results and the patient's current health status, often influence treatment choices.

Before presenting the identification results, we first formally introduce the basic assumptions of the confounded POMDP. Assumption 1 indicates that the future states are independent of the past given current full state and action (S_t, O_t, A_t). Assumption 2 requires that the reward proxy W_t relates to the unobserved state S_t when conditioned on the observed state O_t, but not with the action A_t when conditioned on the current full state (S_t, O_t). Notably, Assumption 2 does not assume the causal relationship between W_t and R_t, thus, the dashed line between W_t and R_t in Figure 1. Assumption 3 requires that the previous observed state, (O_{t-1}, A_{t-1}) does not influence the reward proxy W_t and the reward R_t given current full state and action. It can be easily verified that Assumptions 1–3 are automatically satisfied by the DAG in Figure 1.

Assumption 1 (Markovian). $(S_{t+1}, O_{t+1}) \perp\!\!\!\perp \{S_j, O_j, A_j\}_{1\le j<t} | (S_t, O_t, A_t)$, for $0 \le t \le T$.

Assumption 2 (Reward Proxy). $W_t \perp\!\!\!\perp (A_t, S_{t-1}, O_{t-1}) | (S_t, O_t)$, and $W_t \not\perp\!\!\!\perp S_t | O_t$, for $1 \le t \le T$.

Assumption 3 (Action Proxy). $(O_{t-1}, A_{t-1}) \perp\!\!\!\perp (W_t, R_t) | S_t, O_t, A_t$ for $1 \le t \le T$.

However, we still cannot directly identify the value of target policy only with Assumptions 1–3 by adjusting (S_t, O_t), as S_t is not observable. Thus, we also need the completeness assumption as stated in Assumption 4 to get around the unobserved state S_t.

Assumption 4 (Completeness). (a) For any square-integrable function h, $\mathbb{E}[h(S_t) \mid O_{t-1}, A_{t-1}, O_t, A_t] = 0$ a.s. if and only if $h = 0$ a.s.
(b) For any square-integrable function g, $\mathbb{E}[g(O_{t-1}, A_{t-1}) \mid O_t, W_t, A_t] = 0$ a.s. if and only if $g = 0$ a.s.

Completeness is a commonly made assumption in identification problems, such as instrumental variable identification (Newey and Powell 2003; D'Haultfoeuille 2011), and proximal causal inference (Tchetgen Tchetgen et al. 2020; Cui et al. 2023). Assumption 4 (a) rules out conditional independence between (O_{t-1}, A_{t-1}) and S_t given O_t and A_t, and indicates that the previous state-action pair (O_{t-1}, A_{t-1}) should contain sufficient information from the unobserved state S_t. Assumption 4 (b) ensures the injectivity of the conditional expectation operator. Leveraging Picard's Theorem (Kress, Maz'ya, and Kozlov 1989), the existence of bridge functions within a contextual bandit setting can be established (Miao, Geng, and Tchetgen Tchetgen 2018).

Based on Assumptions 1–4, we generalize the original result to an infinite horizon setting. Define the Q-bridge function and V-bridge function of the target policy π as follows:

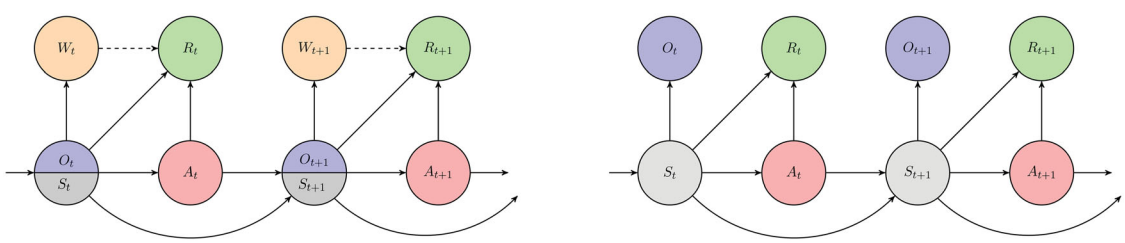

Figure 1. DAG of the proposed confounded POMDP (left) and classical POMDP (right).

$$\mathbb{E}[Q^\pi(O_t, W_t, A_t)|O_t, S_t, A_t] = \mathbb{E}_\pi[\sum_{k=0}^{\infty} \gamma^k R_{t+k}|O_t, S_t, A_t], \tag{3.1}$$

$$\mathbb{E}[V^\pi(O_t, W_t)|O_t, S_t] = \mathbb{E}_\pi[\sum_{k=0}^{\infty} \gamma^k R_{t+k}|O_t, S_t]. \tag{3.2}$$

Therefore, it is obvious that $V^\pi(O_t, W_t) = \int_{a\in\mathcal{A}} \pi(a|O_t)Q^\pi(O_t, W_t, a)da$. If there exists V^π that satisfy (3.2), then the value of target policy π can be identified by

$$J(\pi) = \mathbb{E}[V^\pi(O_0, W_0)].$$

Notice that bridge functions defined in (3.1) and (3.2) are not necessarily unique, but we can uniquely identify the policy value $J(\pi)$ based on any of them. We formally present the identification result in Theorem 3.1.

Theorem 3.1. (Identification) For a confounded POMDP model whose variables satisfy Assumptions 1–4 and some regularity assumptions in Assumption 11 in supplementary materials, there always exist Q-bridges and V-bridges satisfying (3.1) and (3.2), respectively. Additionally, one particular Q-bridge and V-bridge can be obtained by solving the following equation

$$\mathbb{E}[Q^\pi(O_t, W_t, A_t) - R_t - \gamma V^\pi(O_{t+1}, W_{t+1})|O_{t-1}, A_{t-1}, O_t, A_t] = 0. \tag{3.3}$$

Theorem 3.1 guarantees the existence of both V-bridges and Q-bridges. Additionally, the identification equation (3.3) addresses the issue of the unobserved state S_t by conditioning on previous state-action pair (O_{t-1}, A_{t-1}), which forms the basis for estimating bridge functions and eventually estimate the policy value of the target policy $J(\pi)$.

3.2. Minimax Estimation

In this section, we discuss how to estimate the bridge functions using the pre-collected dataset $\mathcal{D}_n = \{O_0^i, W_0^i, A_0^i, R_0^i, O_1^i, \ldots, O_T^i, W_T^i, A_T^i, R_T^i, O_{T+1}^i, W_{T+1}^i\}_{i=1}^n$, which consists of n iid copies of the observable trajectory $(O_t, W_t, A_t, R_t)_{t=1}^T$. Based on the identification equation (3.3), we have for the target policy π and any function f,

$$\mathbb{E}\Big[\Big(Q^\pi(O_t, W_t, A_t) - R_t - \gamma V^\pi(O_{t+1}, W_{t+1})\Big) \times f(O_{t-1}, A_{t-1}, O_t, A_t)\Big] = 0.$$

We denote

$$L_\pi(q,f) = \Big[q(O_t, W_t, A_t) - R_t - \gamma \times \int_{a\in\mathcal{A}} \pi(a|O_{t+1})q(O_{t+1}, W_{t+1}, a)da\Big]f(O_{t-1}, A_{t-1}, O_t, A_t),$$

thus, $\mathbb{E}[L_\pi(Q^\pi, f)] = 0$ for any $f \in \mathcal{F}$. Such observation directly leads us to the minimax estimator

$$\tilde{Q}^\pi = \arg\min_{q\in\mathcal{Q}} \max_{f\in\mathcal{F}} \mathbb{E}[L_\pi(q,f)], \tag{3.4}$$

where we use the function class $\mathcal{Q}$ to model the Q-bridge function, the function class $\mathcal{F}$ to model the critic function f. The corresponding finite-sample estimator is then

$$\hat{Q}^\pi = \arg\min_{q\in\mathcal{Q}} \max_{f\in\mathcal{F}} \mathbb{E}_\mathcal{D}[L_\pi(q,f)] + \lambda_n h_1^2(q) - \frac{1}{2}\mathbb{E}_\mathcal{D}[f^2] - \mu_n h_2^2(f), \tag{3.5}$$

where $\mathbb{E}_\mathcal{D}$ denotes the sample average over all observed tuples $\{O_{t-1}, A_{t-1}, O_t, W_t, A_t, R_t, O_{t+1}, W_{t+1}\}$, $h_1 : \mathcal{Q} \to \mathbb{R}^+, h_2 : \mathcal{F} \to \mathbb{R}^+$ are two regularizers and μ_n, λ_n are tuning parameters.

We observe that $\hat{Q}^\pi$ acts as a penalized estimator for $\tilde{Q}^\pi$. The term $\frac{1}{2}\mathbb{E}_\mathcal{D}[f^2]$ serves as an L_2 regularizer for the function class $\mathcal{F}$, which has been previously explored in the context of reinforcement learning literature (Antos, Szepesvári, and Munos 2008; Hoffman et al. 2011), as well as in the broader domain of minimax estimation problems (Dikkala et al. 2020). The component $\lambda_n h_1^2(q)$ aims to strike a balance between the model's fit regarding the estimated Bellman error and the complexity of the estimated Q-function. Similarly, $\mu_n h_2^2(f)$ is deployed to mitigate overfitting, especially when the function class $\mathcal{F}$ exhibits complexity. The estimated policy value can subsequently be calculated using the following equation,

$$\hat{J}(\pi) = \mathbb{E}_\mathcal{D}\left[\int_{a\in\mathcal{A}} \pi(a|O_0)\hat{Q}^\pi(O_0, W_0, a)da\right]. \tag{3.6}$$

The minimax optimization of (3.5) provides a clear direction to estimate the Q-bridge. In practice, we can use linear basis functions, neural networks, random forests and reproducing kernel Hilbert spaces (RKHSs), etc., to parameterize q and f, and get the estimated $\hat{Q}^\pi$. However, directly solving the minimax optimization problem can be unstable due to its inherent complexities. Furthermore, representing f within an arbitrary function class $\mathcal{F}$ poses additional intractability. Previous work by Uehara, Huang, and Jiang (2020) and Shi et al. (2022) proposed using a linear basis function representation, which yields a closed-form solution for the inner maximization problem in (3.5). Additionally, Dikkala et al. (2020) demonstrated that the minimax problem can be reduced to classification and regression oracles, enabling the use of random forests as oracles. Alternatively, we establish the continuity invariance between the reward function and the optimal critic function $f^*(\cdot)$ in Theorem 3.2, which motivates us to consider kernel representation for $\mathcal{F}$.

Theorem 3.2. Suppose $\mathcal{F} \in L^2(C_0)$, and we define the optimal critic function as $f^* = \arg\max_{f\in\mathcal{F}} \mathbb{E}[L_\pi(q,f)]$. Let $\mathbb{C}(\mathcal{O} \times \mathcal{A} \times \mathcal{O} \times \mathcal{A})$ be all continuous functions on $\mathcal{O} \times \mathcal{A} \times \mathcal{O} \times \mathcal{A}$. For any $(o^-, a^-, o, a) \in \mathcal{O} \times \mathcal{A} \times \mathcal{O} \times \mathcal{A}$ and $s \in \mathcal{S}$, the optimal critic function $f^*(O_{t-1}, A_{t-1}, O_t, A_t) \in \mathcal{F} \cap \mathbb{C}(\mathcal{O} \times \mathcal{A} \times \mathcal{O} \times \mathcal{A})$ is unique if the reward function $R(s, o, a)$ and the transition kernel $\mathbf{P}(s^+, o^+|s, o, a)$ are continuous over (s, o, a), and the density of the target policy π is continuous over $\mathcal{O} \times \mathcal{A}$.

Theorem 3.2 demonstrates that, provided the reward function and the density of the target policy are continuous, which is widely held in real-world scenarios, the optimal $f^*(\cdot)$ remains continuous. Meanwhile, for a positive definite kernel K, a bounded RKHS denoted as $H_{\text{RKHS}}(C_0) := \{f \in H_{\text{RKHS}} :$

$\|f\|^2_{\mathcal{H}_K} \leq C_0\}$ enjoys a diminishing approximation error to any continuous function class as $C_0 \to \infty$ (Bach 2017). Given this observation and the previously mentioned continuity invariance, we propose to represent the critic function within a bounded RKHS. We further show that by kernel representation, the original minimax optimization problem in (3.4) can be decoupled into a single stage minimization problem in Theorem 3.3.

Theorem 3.3. Suppose $\mathcal{F}$ belongs to a bounded reproducing kernel Hilbert space (RKHS), that is, $\mathcal{F} = \{f(o^-, a^-, o, a); f \in \mathcal{H}_K, \|f\|^2_{\mathcal{H}_K} \leq C_0\}$, then the original minimax problem defined in (3.4) can be decoupled into the following minimization problem

$$\tilde{Q}^\pi = \arg\min_{q\in\mathcal{Q}} \mathbb{E}\Big[\Big(q(O_t, W_t, A_t) - R_t - \gamma \times \int_{a\in\mathcal{A}} \pi(a|O_{t+1}) q(O_{t+1}, W_{t+1}, a) da\Big) \tag{3.7}$$
$$K(\{O_{t-1}, A_{t-1}, O_t, A_t\}; \{\bar{O}_{t-1}, \bar{A}_{t-1}, \bar{O}_t, \bar{A}_t\})\Big(q(\bar{O}_t, \bar{W}_t, \bar{A}_t) - \bar{R}_t - \gamma \int_{a\in\mathcal{A}} \pi(a|\bar{O}_{t+1}) q(\bar{O}_{t+1}, \bar{W}_{t+1}, a) da\Big)\Big],$$

where $(\bar{O}_{t-1}, \bar{A}_{t-1}, \bar{O}_t, \bar{W}_t, \bar{A}_t, \bar{O}_{t+1}, \bar{W}_{t+1})$ is an independent copy of the transition pair $(O_{t-1}, A_{t-1}, O_t, W_t, A_t, O_{t+1}, W_{t+1})$.

Theorem 3.3 essentially transforms the minimax problem presented in (3.4) into a single-stage minimization problem through its kernel representation, offering a direct approach to optimization. We note that methods have been developed to directly solve the sample version of (3.7) in both MDP (Uehara, Huang, and Jiang 2020) and POMDP setting (Shi et al. 2022). However, in the context of batch reinforcement learning especially with continuous action space, optimization procedure may be unstable due to limited data without appropriate regularization. To address this, we additionally demonstrate that the finite-sample estimator, as formulated in (3.5), also enjoys a closed-form solution for its inner-maximization part.

Theorem 3.4. Suppose $\mathcal{F} = \{f(o^-, a^-, o, a); f \in \mathcal{H}_K\}$, and $h_2^2(f) := \|f\|^2_{\mathcal{H}_K}$ denotes the kernel norm of f, then the finite-sample minimax problem as shown in (3.5) can be decoupled into the following minimization problem

$$\hat{Q}^\pi := \arg\min_{q\in\mathcal{Q}} \mathcal{L}(q) = \arg\min_{q\in\mathcal{Q}} \Phi^\top(q) K_{nT}^{1/2}\Big[\frac{1}{2nT\mu_n} K_{nT} + I\Big]^{-1} \times K_{nT}^{1/2}\Phi(q) + \lambda_n h_1^2(q), \tag{3.8}$$

where $K_{nT} = K\Big(\{O^i_{t-1}, A^i_{t-1}, O^i_t, A^i_t\}; \{O^{i'}_{t'-1}, A^{i'}_{t'-1}, O^{i'}_{t'}, A^{i'}_{t'}\}\Big)$ is the sample kernel matrix, $\Phi(q) = [\delta^1_{1,\pi}(q), \delta^1_{2,\pi}(q), \ldots, \delta^1_{T,\pi}(q), \delta^2_{1,\pi}(q), \ldots, \delta^n_{T-1,\pi}(q), \delta^n_{T,\pi}(q)]^\top \in \mathbb{R}^{nT}$, and $\delta^i_{t,\pi}(q) = q(O^i_t, W^i_t, A^i_t) - R^i_t - \gamma \int_{a\in\mathcal{A}} \pi(a|O^i_{t+1}) q(O^i_{t+1}, W^i_{t+1}, a) da$.

Notice that by including the kernel norm $h_2^2(f) := \|f\|^2_{\mathcal{H}_K}$ as the penalty term, we drop the boundedness constraint on the function class $\mathcal{F}$ for the finite-sample estimator. Additionally, when selecting μ_n, λ_n such that $\mu_n, \lambda_n, \frac{1}{n\mu_n} \to 0$, estimating (3.8) converges to the form (3.7). Thus, (3.8) can be considered as a regularized variant of (3.7), and provides a clear path in estimating the Q-bridge function. By parameterizing the Q-bridge function with parameter θ using tools such as linear basis functions, RKHS, neural networks among others, we introduce an SGD-based algorithm as outlined in Algorithm 1 to determine the estimated $\hat{Q}^\pi$.

Algorithm 1 Off-Policy Evaluation

1: **Input** observed transition pairs data $\{O_{t-1}, A_{t-1}, O_t, W_t, A_t, R_t, O_{t+1}, W_{t+1}\}_{t=1}^n$, target policy π.
2: **Initialize** the parameters of interests $\theta = \theta^{(0)}$, the mini-batch size n_0, the learning rate α_0, the kernel bandwidth bw_0, and the stopping criterion ε.
3: **For iterations** $j = 1$ to k
4: Randomly sample a mini-batch $\{O_{t-1}, A_{t-1}, O_t, W_t, A_t, R_t, O_{t+1}, W_{t+1}\}_{t=1}^{n_0}$.
5: Decay the learning rate $\alpha_j = \mathcal{O}(j^{-1/2})$.
6: Compute stochastic gradients with respect to θ, $\nabla_\theta \mathcal{L}(q)$, in (3.8).
7: Update the parameters of interest as $\theta^{(j)} \leftarrow \theta^{(j-1)} - \alpha_j \nabla_\theta \mathcal{L}(q)$.
8: Stop if $\|\theta^{(j)} - \theta^{(j-1)}\| \leq \varepsilon$.
9: **Return** $\widehat{\theta} \leftarrow \theta^{(j)}$.

3.3. Policy Learning

Section 3.2 outlines the methodology for estimating the Q-bridge. Following this, the value of a specific target policy π can be determined by $\mathbb{E}[V^\pi(O_0, W_0)] = \mathbb{E}\left[\int_{a\in\mathcal{A}} \pi(a|O_0) Q^\pi(O_0, W_0, a) da\right]$. It is important to note that the induced estimator $\hat{Q}^\pi$ is unbiased and exhibits a convergence rate of $O(n^{-1/2})$, as demonstrated in Section 4. Consequently, a natural approach for identifying the optimal policy π^* is to search for the policy π that maximizes the estimated policy value, that is

$$\pi^* = \arg\max_{\pi\in\Pi} J(\pi) = \arg\max_{\pi\in\Pi} \mathbb{E}\left[\int_{a\in\mathcal{A}} \pi(a|O_0) Q^\pi(O_0, W_0, a) da\right]. \tag{3.9}$$

One may consider policy-iteration-type algorithms that alternate between policy iteration and improvement to solve (3.9). However, the reward proxies W_t in our framework may include variables that are only observed after the action is taken. As a result, the optimal policy induced by such algorithms, expressed as $\pi(o, w) = \arg\max_{a\in\mathcal{A}} Q^\pi(o, w, a)$ may not be directly applicable in the context of a confounded POMDP, where w is unavailable at the time of action selection. Thus, we propose to parameterize the policy distribution with $\pi(\zeta)$ to capture the learning process leading to the induced optimal policy, and the parameters corresponding to the optimal policy can then be obtained by solving the following optimization problem,

Algorithm 2 Off-Policy Learning

1: **Input** observed transition pairs data $\{O_{t-1}, A_{t-1}, O_t, W_t, A_t, R_t, O_{t+1}, W_{t+1}\}_{t=1}^n$.
2: **Initialize** the parameters of interests $\zeta = \zeta^{(0)}$, $\theta = \theta^{(0)}$, the mini-batch size n_0, the learning rate α_0, β_0, the kernel bandwidth bw_0, and the stopping criterion ε.
3: **For iterations** $j = 1$ to k
4: Implement Algorithm 1 to get $\theta^{(j)}$.
5: Compute the gradient with respect to ζ, $\nabla_\zeta L_\pi(\zeta^{(j-1)}, \theta^{(j)})$ in (3.11).
6: Decay the learning rate $\beta_j = \mathcal{O}(j^{-1/2})$.
7: Update the parameter $\zeta^{(j)} \leftarrow \zeta^{(j-1)} + \beta_j \nabla_\zeta \widehat{L_\pi}$.
8: **Return** $\widehat{\zeta} \leftarrow \zeta^{(j)}$.

$$\hat{\pi}^*(o, a; \zeta) = \arg\max_{\pi \in \Pi} \hat{J}(\pi) = \arg\max_{\pi \in \Pi} \mathbb{E}_{\mathcal{D}}\Big[\int_{a \in \mathcal{A}} \pi(a|O_0; \zeta) \hat{Q}^\pi(O_0, W_0, a; \theta) da\Big]. \tag{3.10}$$

Similar to the Q-bridge, practical parameterization of distribution parameters, such as the mean and variance of the truncated normal distribution or the (α, β) parameters for a beta distribution can be established using linear basis, neural networks, RKHS or others.

To effectively tackle the optimization problem defined in (3.10), we propose an algorithm that facilitates iterative updates to the parameters (ζ, θ). Notice that for a predetermined ζ, the corresponding Q-bridge $Q^{\pi(\zeta)}(o, w, a; \theta)$ can be fully determined by ζ, thus, θ can be considered as a function of ζ. Consequently, to solve the optimization problem in (3.10), we start with an initial policy parameter $\zeta^{(0)}$, thereby defining the policy $\pi(\zeta^{(0)})$. Following the specification of initial value, we apply Algorithm 1 to estimate $\theta^{(0)}$ and the associated Q-bridge function $\hat{Q}^{\pi(\zeta^{(0)})}(O, W, A; \theta(\zeta^{(0)}))$. Subsequently, the policy gradient with respect to ζ from (3.10) is calculated as

$$\nabla_\zeta L_\pi(\theta, \zeta) = \mathbb{E}_{\mathcal{D}}[\nabla_\zeta \int_{a \in \mathcal{A}} \pi(a|O_0; \zeta) Q^\pi(O_0, W_0, a; \theta(\zeta)) da], \tag{3.11}$$

$$= \mathbb{E}_{\mathcal{D}}\Big[\int_{a \in \mathcal{A}} \Big\{\nabla_\zeta \pi(a|O_0; \zeta) Q^\pi(O_0, W_0, a; \theta(\zeta)) + \pi(a|O_0; \zeta) \nabla_\zeta Q^\pi(O_0, W_0, a; \theta(\zeta))\Big\} da\Big].$$

Gradient ascent is employed to update $\zeta^{(0)}$, with the objective of obtaining a policy with a larger estimated value. This procedure is repeated to update the parameters (ζ, θ) iteratively until convergence is attained, at which point (ζ, θ) are considered to approximate the optimal policy and its corresponding Q-bridge function. This iterative update mechanism is outlined in Algorithm 2. We note that this policy learning mechanism is broadly applicable to identification results based on different causal graphs. This can be achieved by parameterizing the policy class and replacing the policy evaluation step in Algorithm 2 with methods designed for alternative settings. This approach offers significant computational advantages when dealing with continuous action spaces, as traditional policy iteration algorithms typically require solving an additional optimization problem $\max_{a \in \mathcal{A}} Q^\pi(o, w, a)$ at each iteration.

4. Theoretical Results

In this section, we first derive the global rate of convergence for $\hat{Q}^\pi$ in (3.5), and the finite-sample error bound for the estimated policy value $\hat{J}(\pi)$ in (3.6). We then extend the results to the estimated optimal policy, where we derive the regret bound for $\hat{\pi}^*$ defined in (3.10). We note that while we use kernel representation to obtain the closed-form solution for the inner-maximization problem as demonstrated in (3.7) and (3.8), our theoretical results are based on the general form of (3.5). To simplify notation, we denote the L_2 norm with respect to the average state-action distribution in the trajectory $\mathcal{D}$ as $\|f\|^2 = \mathbb{E}\Big[\frac{1}{T}\sum_{t=1}^T f^2(\cdot)\Big]$, and the Bellman operator $\mathcal{T}_\pi$ with respect to the target policy π as $\mathcal{T}_\pi(s, o, a; q) = \mathbb{E}\Big[R_{t+1} + \gamma \int_{a \in \mathcal{A}} \pi(a'|O_{t+1}) q(O_{t+1}, W_{t+1}, a') da \mid S_t = s, O_t = o, A_t = a\Big]$. Before presenting our main results, we first state several standard assumptions on policy class Π, reward function R_t, and function class $\mathcal{Q}$ and $\mathcal{F}$.

Assumption 5. The target policy class $\Pi = \{\pi_\zeta : \zeta \in Z \subset \mathbb{R}^p\}$, satisfies:

(i) $H \subset \mathbb{R}^p$ is compact and let $\text{diam}(Z) = \sup\{\|\zeta_1 - \zeta_2\|_2 : \zeta_1, \zeta_2 \in H\}$.

(ii) There exists $L_Z > 0$, such that for $\zeta_1, \zeta_2 \in H$ and for all $(o, a) \in \mathcal{O} \times \mathcal{A}$, the following holds

$$|\pi_{\zeta_1}(a|o) - \pi_{\zeta_2}(a|o)| \le L_Z \|\zeta_1 - \zeta_2\|_2.$$

Assumption 6. (i) The reward is uniformly bounded: $|R_t| \le R_{\max}$ for all $t \ge 0$.

(ii) The function class $\mathcal{Q}$ satisfies $\|q\|_\infty \le q_{\max}$ for all $q \in \mathcal{Q}$, and $Q^\pi \in \mathcal{Q}$ for all $\pi \in \Pi$.

Assumption 7. The function class $\mathcal{F}$ satisfies (i) $0 \in \mathcal{F}$; (ii) $\|f\|_\infty \le f_{\max}$ for all $f \in \mathcal{F}$; (iii) $\kappa = \inf\{\|\tilde{f}\| : \|\mathcal{T}_\pi(s, o, a; q) - q(o, w, a)\| = 1, q \in \mathcal{Q}, \pi \in \Pi\} > 0$, where $\tilde{f} = \arg\max_f \mathbb{E}[L_\pi(q, f) - \frac{1}{2}f^2]$.

Assumption 5 is used to control the complexity of policy class, which are commonly assumed in policy learning problems (Liao et al. 2022; Wang and Zou 2022). Assumption 6 is a regular assumptions to impose boundedness condition on reward and function class $\mathcal{Q}$ (Antos, Szepesvári, and Munos 2008; Bennett et al. 2021). Assumption 7 similarly ensures the boundedness of function class $\mathcal{F}$. Additionally, the value of κ measures how well the function class $\mathcal{F}$ approximates the Bellman error for all $q \in \mathcal{Q}, \pi \in \Pi$. This concept has been explored as the well-posedness condition within the field of off-policy evaluation (Chen and Qi 2022; Miao, Qi, and Zhang 2022). A strictly positive value of κ indicates a substantial overlap between the behavior and target policies in terms of the state-action distribution, ensuring the identification of the true policy value $J(\pi)$.

Assumption 8. (i) The regularization functionals, h_1 and h_2, are pseudo norms and induced by the inner products, $h_1(\cdot, \cdot)$ and

$h_2(\cdot,\cdot)$, respectively. There exist constants, C_1 and C_2, such that $h_2(\tilde{f}(q)) \le C_1 + C_2 h_1(q)$ holds for all $q \in \mathcal{Q}$.

(ii) Let $\mathcal{Q}_M = \{c + q : |c| \le R_{\max}, q \in \mathcal{Q}, h_1(q) \le M\}$, and $\mathcal{F}_M = \{f : f \in \mathcal{F}, h_2(f) \le M\}$. There exists constant C_3 and $\alpha \in (0,1)$ such that for any $\epsilon, M > 0$,

$$\max\left\{\log N(\varepsilon, \mathcal{F}_M, ||\cdot||_\infty), \log N(\varepsilon, \mathcal{Q}_M, ||\cdot||_\infty)\right\} \le C_3\left(\frac{M}{\varepsilon}\right)^{2\alpha}.$$

Assumption 8 characterizes the complexity of the function class $\mathcal{Q}$ and $\mathcal{F}$. The condition that the regularizers be pseudo-norms is satisfied for common function classes such as RKHS and Sobolev spaces (Geer 2000; Thomas and Brunskill 2016). The upper bound on h_2 is realistic when the transition kernel is sufficiently smooth (Farahm et al. 2016). We use a common $\alpha \in (0,1)$ for both $\mathcal{Q}$ and $\mathcal{F}$ to simplify the proof.

Theorem 4.1. Suppose the target policy $\pi \in \Pi$, and let $\hat{Q}^\pi$ be the estimator defined in (3.5). Suppose Assumptions 1–8 hold, and the tuning parameter satisfy $\frac{1}{\tau} n^{-\frac{1}{1+\alpha}} \le \mu_n \le \tau\lambda_n$ for some constant $\tau > 0$. Then the following bounds hold with probability at least $1-\delta$ where $\delta \in (0,1)$,

$$\|\mathcal{T}_\pi(\cdot,\cdot,\cdot;\hat{Q}^\pi) - \hat{Q}^\pi\|^2 \lesssim \frac{1}{\kappa^2}\Big[\lambda_n\left[1 + h_1^2(Q^\pi)\right]\left\{1 + \log(1/\delta)\right\}]\Big],$$

where the leading constant depend only on $(\tau, R_{\max}, q_{\max}, f_{\max}, C_1, C_2, C_3, \alpha, \gamma)$.

In Lemma 4 from the supplementary materials, we have proved that $\|\hat{Q}^\pi - Q^\pi\|^2 \lesssim \|\mathcal{T}_\pi(\cdot,\cdot,\cdot;\hat{Q}^\pi) - \hat{Q}^\pi\|^2$. Consequently, Theorem 4.1 demonstrates that $\hat{Q}^\pi$ serves as a consistent estimator for Q^π, provided that $\lambda_n = o_P(1)$. When the tuning parameter is selected such that $\lambda_n \asymp \mu_n$ and $\lambda_n \asymp n^{-1/(1+\alpha)}$, we achieve the optimal convergence rate of the Bellman error at $\hat{Q}^\pi$, which is $O_P(n^{-1/(1+\alpha)})$. Before delving into the finite-sample error bound for $\hat{J}(\pi)$, we first introduce some additional notations.

We define the discounted state-action visitation of target policy π as $d^\pi(o,s,a) = (1-\gamma)\sum_{t=0}^\infty \gamma^t d_t^\pi(o,s,a)$, where d_t^π is the density of the state-action pair at tth time point under the target policy π, and denote the average density over T decision times under the behavior policy as $\bar{d}_T(o,s,a)$. We further define the direction function $e^\pi(o^-,a^-,o,a)$ by

$$\mathbb{E}[e^\pi(o^-,a^-,o,a)|s,o,a] = \frac{d^\pi(o,s,a)}{\bar{d}_T(o,s,a)}. \tag{4.1}$$

The direction function e^π is used to control the bias $|\hat{J}(\pi) - J(\pi)|$ caused by the penalization of Q-bridge in (3.5). As demonstrated in Uehara, Huang, and Jiang (2020); Shi et al. (2022), if we define

$$g^\pi(q) = \frac{1}{T}\sum_{t=1}^T\Big\{q(O_t,W_t,A_t) - \gamma\Big(\int_{a\in\mathcal{A}}\pi(a|O_{t+1})q(O_{t+1},W_{t+1},a)da\Big)\Big\}.$$

$$e^\pi(O_{t-1},A_{t-1},O_t,A_t) - (1-\gamma)\int_{a\in\mathcal{A}}\pi(a|O_0)q(O_0,W_0,a)da, \tag{4.2}$$

then the direction function e^π enjoys the property that $\mathbb{E}[g^\pi(q)] = 0$. Next, we define $\mathbb{E}[\xi^\pi(O_t,W_t,A_t)|S_t,O_t,A_t] = \mathbb{E}[\sum_{k=0}^\infty \gamma^k \frac{d^\pi(O_{t+k},S_{t+k},A_{t+k})}{\bar{d}_T(O_{t+k},S_{t+k},A_{t+k})}]$.

Note that ξ^π has a similar structure to the Q-bridge function, where the "reward" at time t is defined as $\frac{d^\pi(O_t,S_t,A_t)}{\bar{d}_T(O_t,S_t,A_t)}$. Similar to the Q-bridge, ξ^π satisfies the following Bellman-like equation as demonstrated in Theorem 3.1:

$$\mathbb{E}\Big[\xi^\pi(O_t,W_t,A_t) - e^\pi(O_{t-1},A_{t-1},O_t,A_t) + \gamma\int_{a'\in\mathcal{A}}\pi(a'|O_{t+1}) \times \xi^\pi(O_{t+1},W_{t+1},a')da' \mid O_{t-1},A_{t-1},O_t,A_t\Big] = 0.$$

We make the following smoothness assumption about e^π and ξ^π.

Assumption 9. The direction function $e^\pi \in \mathcal{F}$, and $\xi^\pi \in \mathcal{Q}$ for all $\pi \in \Pi$.

The condition on direction function e^π is to ensure that e^π is sufficiently smooth, which will be used to show that the bias of the estimator in (3.5) decreases sufficiently fast to 0. This assumption holds when the function class $\mathcal{F}$ is rich enough, and the mapping from policy parameters to the induced distribution over states and actions is continuous and small perturbations in policy parameters would not lead to large or discontinuous changes. The assumption on ξ^π is analog to the assumptions used in partially linear regression (Geer 2000). The counterpart of $\xi^\pi \in \mathcal{Q}$ in partially linear regression problem, $Y = g(Z) + X^\top\beta + \epsilon$ is the standard assumption that $\mathbb{E}[X|Z = \cdot] \in \mathcal{G}$, where $\mathcal{G}$ is the function class to model $g(Z)$. The last assumption concerns the coverage of the collected dataset.

Assumption 10. (a) There exist positive constants $p_{1,\min}, p_{2,\min}$, and $p_{1,\max}$, such that the visitation density $\bar{d}_T, d_0^\pi(o,s,a)$ satisfy $p_{1,\min} \le \bar{d}_T, d_0^\pi \le p_{1,\max}$ for all $\pi \in \Pi$, and the behavior policy $\pi^b \ge p_{2,\min}$ for every $(o,s,a) \in \mathcal{O}\times\mathcal{S}\times\mathcal{A}$.

(b) The target policy π is absolute continuous with respect to behavior policy π^b for all $\pi \in \Pi$, and $\mathcal{P}^\pi(o',s',w',a'|o,s,w,a) \le p_{2,\max}$ for some positive constant $p_{2,\max}$, where $\mathcal{P}^\pi$ denotes the 1-step visitation density induced by target policy π.

We define $p_{\min} = \min\{p_{1,\min}, p_{2,\min}\}, p_{\max} = \max\{p_{1,\max}, p_{2,\max}\}$. Assumption 10 (a) is frequently referred as the coverage assumption in RL literature (Precup 2000; Kallus and Uehara 2020), which guarantees the collected offline data sufficiently covers the entire state-action space. Importantly, in the presence of unmeasured confounders, this coverage requirement applies to both the observed and unobserved components of the state space, so that policy evaluation and learning remain feasible even when part of the state is unobserved. Similar to traditional literature on causal inference, a random policy would satisfy this assumption. Assumption 10(b) imposes one mild condition on the target policy, which essentially states that the collected batch data is

able to identify the true value of target policy. We now analyze the performance error between the finite-sample estimator and the true policy value.

Theorem 4.2. Suppose the assumptions in Theorem 4.1 hold. Additionally, suppose Assumptions 9 and 10 hold, and $\lambda_n = o(n^{-1/2})$. Then, with probability at least $1-\delta$ where $\delta \in (0,1)$, the estimation error from (3.5) is upper bounded by

$$|\hat{J}(\pi) - J(\pi)| \leq \frac{2[(1+\gamma)q_{\max} + R_{\max}]f_{\max}}{1-\gamma}\sqrt{\frac{\log(2/\delta)}{2n}} + o_P(n^{-1/2}).$$

From Theorem 4.2, it is evident that the proposed estimator, as defined in (3.5), achieves a convergence rate of $O(n^{-1/2})$. This finite-sample error bound effectively extends the result from Shi et al. (2022) to encompass a wider range of function classes, while still preserving the optimal convergence rate of $O(n^{-1/2})$. Moreover, Theorem 4.2 not only details the convergence rate for a specified target policy π, but also sets the groundwork for extending this outcome to the finite-sample regret bound applicable to policy learning.

Proposition 4.1. Let $\hat{Q}^\pi$ be the estimator defined in (3.5). Suppose Assumptions 1–8 hold, and the tuning parameter satisfy $\frac{1}{\tau}n^{-\frac{1}{1+\alpha}} \leq \mu_n \leq \tau\lambda_n$ for some constant $\tau > 0$. Then the following bound holds with probability at least $1-\delta$ where $\delta \in (0,1)$, for all $\pi \in \Pi$,

$$\|\mathcal{T}_\pi(\cdot,\cdot,\cdot;\hat{Q}^\pi) - \hat{Q}^\pi\|^2 \lesssim \frac{p}{\kappa^2}\left[\lambda_n\left[1 + h_1^2(Q^\pi)\right]\left\{1 + \log(1/\delta)\right\}\right],$$

where the leading constant depend only on $(\tau, R_{\max}, q_{\max}, f_{\max}, C_1, C_2, C_3, \mathrm{diam}(Z), L_Z, \alpha, \gamma)$.

Proposition 4.1 extends the error bound in Theorem 4.1 to all $\pi \in \Pi$ with additional efforts in controlling the complexity of policy class Π. This demonstrates that $\hat{Q}^\pi$ is a consistent estimator for any π in Π. Building on this, we analyze the suboptimality of the estimated optimal policy $\hat{\pi}^*$.

Theorem 4.3. Suppose $\hat{\pi}^*$ and π^* are defined in (3.9) and (3.10), respectively, and the assumptions in Theorem 4.1 hold. Then with probability at least $1-\delta$ where $\delta \in (0,1)$, we have

$$J(\pi^*) - J(\hat{\pi}^*) \leq C(\delta)(p^{1/2}n^{-1/2}) + o_P(p^{1/2}n^{-1/2}),$$

where $C(\delta)$ is a function of $(\tau, R_{\max}, q_{\max}, f_{\max}, p_{\max}, C_1, C_2, C_3, \mathrm{diam}(Z), L_Z, \alpha, \gamma)$. The definition of $C(\delta)$ and the constants can be found in the supplementary materials.

Theorem 4.3 is based on the critical observation that $J(\pi^*) - J(\hat{\pi}^*) \leq (J(\pi^*) - \hat{J}(\pi^*)) + (\hat{J}(\pi^*) - \hat{J}(\hat{\pi}^*)) + (\hat{J}(\hat{\pi}^*) - J(\hat{\pi}^*))$, where the error bounds for the first and last terms can be established from the uniform error bound on $|\hat{J}(\pi) - J(\pi)|$ for all $\pi \in \Pi$. Recall that p is the number of parameters in the policy, n is the number of iid trajectories in the collected dataset, and the regret of the estimated optimal policy is $O(p^{1/2}n^{-1/2})$. The first term represents the regret of an estimated policy as if the q-bridge were known beforehand, and the second term arises from the estimation error of the q-bridge function. Theorem 4.3 essentially characterizes the regret of the estimated optimal in-class policy without double robustness, achieving the optimal minimax convergence rate of $O(n^{-1/2})$.

5. Simulation Studies

In this section, we evaluate our proposed method through extensive simulation studies. Since our framework introduces the reward-inducing proxy variable and is designed for both continuous state and action spaces, existing methods for POMDPs are not directly applicable (Shi et al. 2022; Miao, Qi, and Zhang 2022; Hong, Qi, and Xu 2023). Therefore, we compare our proposed method to other methods designed for MDP scenarios. For off-policy evaluation, we adopted the approach proposed by Uehara, Huang, and Jiang (2020), considering two baseline scenarios: one treating O_t as the state variable (MDP) and the other using (O_t, W_t) as the state variables (MDPW). For policy learning, we benchmark our method against state-of-the-art baselines, including implicit Q-learning (IQL) (Kostrikov, Nair, and Levine 2021), soft actor-critic (SAC) (Haarnoja et al. 2018) and conservative Q-learning (CQL) (Kumar et al. 2020). Additionally, we included an MDP-based variant of our proposed approach as a baseline for policy learning task, where we employed the same policy class but substituting our policy evaluation step in Algorithm 2 with the MDP-based evaluation technique developed by Uehara, Huang, and Jiang (2020).

For practical implementation, both $Q^\pi(\cdot;\theta)$ and $\pi(\cdot;\zeta)$ need to be parameterized. We parameterize Q^π as a RKHS using second-degree polynomial kernels. Details can be found in the supplementary materials. The function class $\mathcal{F}$ is specified as a Gaussian RKHS where the bandwidth is selected according to the median heuristic by Fukumizu et al. (2009). The penalty terms $h_1(q), h_2(f)$ are selected as the kernel norm of q, f, respectively. Thus, it can be verified that Assumptions 6–8 are automatically satisfied by the above choice.

To determine the values of the tuning parameters (λ_n, μ_n), we employ a k-fold cross-validation approach. For each candidate pair of tuning parameters, we implement Algorithm 2 to search for the optimal policy on the training set and then evaluate the learned optimal policy on the validation set. We select the optimal (λ_n, μ_n) that maximizes the estimated optimal policy value on the validation set, given by: $(\hat{\lambda}_n, \hat{\mu}_n) = \arg\max_{\lambda_n,\mu_n} \frac{1}{k}\sum_{r=1}^k \hat{J}^{(r)}_{\lambda_n,\mu_n}(\hat{\pi}^*)$, where $\hat{J}^{(r)}_{\lambda_n,\mu_n}(\hat{\pi}^*)$ represents the estimated optimal policy value on the rth validation set. The results of the sensitivity analysis are summarized in the supplementary materials.

We conduct numerical experiments on two synthetic environments to evaluate the finite sample performance of our proposed method. In both environments, the state and action spaces are continuous, and the discount factor is set to be $\gamma = 0.9$ for all experiments. For Environment I, we consider a one-dimensional continuous state space and a continuous action space, where the unobserved initial state follows $S_0 \sim \mathrm{Unif}(-0.5, 0.5)$. The observed state O_t is generated according to the additive noise model, that is, $O_t = S_t + \mathcal{N}(0,1)$. The reward proxy, reward function, and state transition are given by $R_t = A_t(O_t - 0.2W_t - 0.8S_t) - 0.8A_t^2$, $W_t = S_t - 0.5O_t + N(0, 0.3^2)$, $S_{t+1} = 0.8O_t - 0.3A_t + N(0, 0.1^2)$. To ensure sufficient coverage of the observed

dataset, we set the behavior policy to be $\mathcal{N}(-S_t/3, 0.4^2)$, with sampled actions clipped to the interval $[-1, 1]$. For Environment II, we consider a five-dimensional state space to evaluate performance of our proposed method in more complex environments. As in the environment I, the action remains continuous and bounded, but the transition dynamics introduce additional structural dependencies. Details of Environment II and the data generation process are provided in the supplementary materials.

For both environments, we conduct policy evaluation by selecting two different target policies, where Target Policy I is chosen to be similar to the behavior policy and Target Policy II is considered to be near-optimal and farther from the behavior policy. The details of each target policy and implementation details are presented in the supplementary materials. We apply Algorithm 1 to estimate the target policy value, and consider the following sample size configurations $(n, T) \in \{(25, 25), (50, 25), (75, 50), (100, 50)\}$. To evaluate the model performance, we generate 1000 independent trajectories under the target policy and define the true policy value as the simulated mean cumulative reward. We compare the relative MSE of our proposed estimator with its MDP and MDPW counterparts. The simulation results are shown in Figure 2.

Figure 2 shows that our proposed method achieves smaller bias in policy evaluation across all settings. This indicates that by leveraging the identification result presented in Theorem 3.1, we can successfully recover the lost information from unobserved variables with the help of action and reward-inducing proxies. Additionally, it can be observed that the bias of the proposed method generally decreases with larger sample sizes, while the bias under MDP method remains the same across different scenarios. This further demonstrates that ignoring the effect of unmeasured confounders leads to a biased estimate.

For policy learning, we consider both Beta and Gaussian policy classes, where the distribution parameters are parameterized by either a linear basis or a neural network. For the neural network parameterization, we use a one-hidden-layer MLP to approximate the distribution parameters in both policy classes. For the linear basis parameterization of Beta distribution, we define the policy class as $\text{Beta}(\text{Softplus}(\zeta_1^\top o), \text{Softplus}(\zeta_2^\top o))$ with actions linearly rescaled to $[-1, 1]$, where $o \in \mathbb{R}^d$ is the state vector and intercept, and $\zeta_1, \zeta_2 \in \mathbb{R}^d$ are parameter vectors to be estimated. Similarly, for the Gaussian policy class, we let $\tanh^{-1}(A) \sim \mathcal{N}(\mu(o), \sigma^2(o)))$ with $(\mu(o), \log(\sigma(o))) := (\zeta_1^\top o, \zeta_2^\top o)$. While the neural network is capable of learning more complex optimal policies, the linear policy class offers greater interpretability and may be preferable in real-world applications. We additionally constraint the policy parameters within a certain range to avoid abnormal distributions. Assumption 5 and 9 are automatically satisfied by such policy classes. Details on the choice of policy classes are provided in the Supplementary Materials.

To evaluate the performance of our proposed method, we apply Algorithm 2 and compare its results against the MDP and MDPW variants of the same approach, as well as against state-of-the-art policy learning algorithms including IQL, SAC, and CQL. To ensure a fair comparison, we employ the identical policy class for all policy parameterizations of the MDP and MDPW counterparts of our proposed method. Since IQL, SAC, and CQL are designed for deep reinforcement learning, we restrict their policy class to a Gaussian neural network. All data are generated using the synthetic environments and behavior policies described in the policy evaluation section, and Table 1 summarizes the results over 50 simulation runs.

Table 1 shows that our proposed method outperforms competing methods across different environments and sample sizes, with comparable variance. This is mainly due to the unbiased estimation of target policies, thus, ensuring that the policy gradients are in the right direction. In comparison, the MDP version of our proposed method may suffer from biased estimation of the target policies, updating the policy in a sub-optimal direction and leading to relatively poor performance. Additionally, Gaussian and Beta policy classes show similar performance under different settings, which demonstrates that our approach remains robust to the choice of policy class. Meanwhile, policy learning methods for MDP settings (SAC, CQL, and IQL) aim at directly solving the Bellman optimality equation. However, the Bellman optimality equation does not hold due to the existence of unmeasured confounders, thus, negatively affecting the performance of these methods. Additionally, our proposed method is capable of achieving stable performance with relatively small sample sizes, which is desirable in many real-world applications.

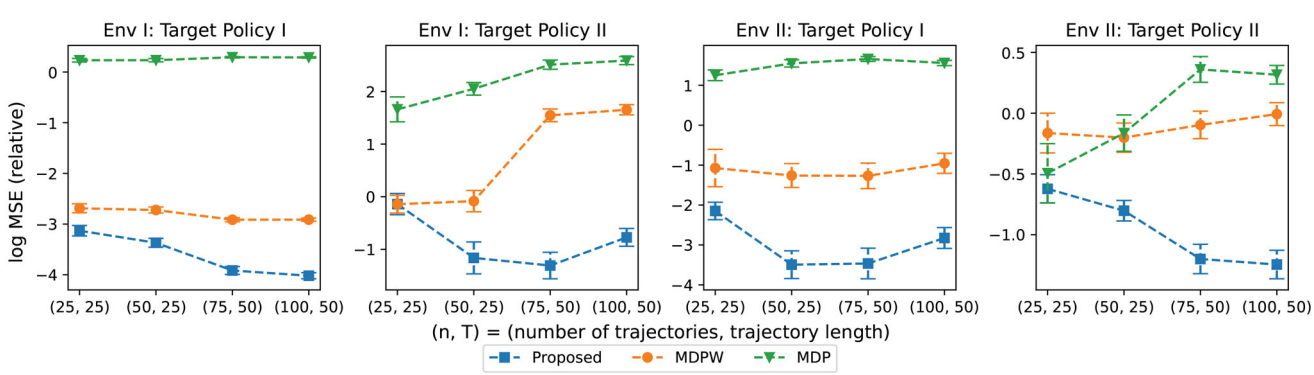

Figure 2. Logarithms of relative MSEs of the proposed, MDPW, and MDP estimators and their associated 95% confidence interval, based on 50 simulation runs, with different choices of (n, T). The definition of relative MSE can be found in the supplementary materials.

Table 1. The mean and standard deviation (in parentheses) of the learned optimal policy value over 50 simulation runs across different environment and sample sizes.

	Environment I				Environment II			
Method/(n, T)	(25, 25)	(50, 25)	(75, 50)	(100, 50)	(25, 25)	(50, 25)	(75, 50)	(100, 50)
Proposed-NN-Gauss	1.63 (0.19)	1.70 (0.16)	1.80 (0.16)	1.77 (0.13)	1.03 (0.61)	1.35 (0.58)	1.38 (0.45)	1.54 (0.48)
Proposed-NN-Beta	1.62 (0.36)	1.68 (0.24)	1.71 (0.13)	1.72 (0.12)	1.12 (0.55)	1.25 (0.46)	1.10 (0.55)	1.27 (0.44)
Proposed-Linear-Gauss	1.66 (0.13)	1.65 (0.09)	1.68 (0.07)	1.71 (0.07)	0.94 (0.59)	1.16 (0.66)	1.10 (0.55)	1.31 (0.58)
Proposed-Linear-Beta	1.54 (0.52)	1.64 (0.35)	1.85 (0.20)	1.94 (0.17)	1.10 (0.51)	1.29 (0.26)	1.35 (0.36)	1.44 (0.29)
MDPW-NN-Gaussian	1.36 (0.29)	1.30 (0.14)	1.41 (0.20)	1.33 (0.36)	0.71 (0.43)	0.83 (0.41)	1.02 (0.37)	1.03 (0.36)
MDPW-NN-Beta	1.27 (0.38)	1.34 (0.31)	0.93 (0.43)	0.95 (0.33)	0.84 (0.45)	0.98 (0.34)	1.08 (0.39)	1.29 (0.41)
MDPW-Linear-Gauss	1.41 (0.19)	1.42 (0.12)	1.41 (0.10)	1.37 (0.10)	0.73 (0.46)	0.86 (0.38)	0.95 (0.32)	1.09 (0.42)
MDPW-Linear-Beta	1.27 (0.14)	1.28 (0.13)	1.02 (0.58)	0.98 (0.51)	0.84 (0.40)	1.05 (0.27)	1.08 (0.26)	1.17 (0.29)
MDPW-SAC	0.72 (1.13)	1.04 (0.34)	1.23 (0.17)	1.26 (0.15)	0.59 (0.25)	0.72 (0.25)	1.05 (0.15)	1.14 (0.13)
MDPW-CQL	1.41 (0.39)	1.44 (0.39)	1.30 (0.32)	1.29 (0.30)	−0.11 (0.24)	0.31 (0.20)	0.89 (0.20)	1.04 (0.20)
MDPW-IQL	0.61 (0.33)	0.90 (0.21)	0.94 (0.16)	0.98 (0.13)	0.15 (0.26)	0.21 (0.23)	0.51 (0.23)	0.40 (0.27)
MDP-NN-Gauss	0.55 (0.39)	0.10 (0.28)	−0.28 (0.10)	−0.18 (0.24)	0.38 (0.35)	0.19 (0.39)	0.27 (0.44)	0.15 (0.39)
MDP-NN-Beta	−0.21 (0.19)	−0.32 (0.10)	−0.24 (0.11)	−0.22 (0.17)	0.25 (0.24)	0.15 (0.17)	0.13 (0.13)	0.13 (0.12)
MDP-Linear-Gaussian	0.86 (0.53)	0.91 (0.48)	1.08 (0.14)	1.04 (0.12)	0.35 (0.42)	0.54 (0.48)	0.42 (0.43)	0.54 (0.44)
MDP-Linear-Beta	0.46 (0.19)	0.38 (0.09)	0.11 (0.15)	0.10 (0.21)	0.40 (0.21)	0.28 (0.16)	0.27 (0.12)	0.27 (0.11)
MDP-SAC	0.22 (0.64)	0.31 (0.34)	0.28 (0.12)	0.32 (0.10)	0.37 (0.17)	0.48 (0.12)	0.51 (0.10)	0.55 (0.08)
MDP-CQL	0.44 (0.35)	0.50 (0.27)	0.41 (0.17)	0.36 (0.18)	0.12 (0.23)	0.31 (0.18)	0.31 (0.17)	0.36 (0.15)
MDP-IQL	0.29 (0.35)	0.57 (0.25)	0.66 (0.19)	0.73 (0.14)	−0.28 (0.16)	−0.14 (0.13)	0.28 (0.13)	0.39 (0.15)

NOTE: The true policy values are obtained from Monte Carlo simulations with $T = 100$, $n = 1000$. The behavior policy value for Environment I and Environment II are −1.39 and −2.87, respectively.

In terms of the comparison between the neural network and linear policy classes, neural networks show better performance due to the richness of the function class, but the linear basis is capable of achieving comparable performance with fewer and more interpretable parameters, and thus may be of more interest in real-world applications, as demonstrated in Section 6.

6. Application to Pairfam Dataset

We applied our proposed method to the Panel Analysis of Intimate Relationships and Family Dynamics (Pairfam) dataset (Brüderl et al. 2023). Initiated in 2008, Pairfam is a comprehensive longitudinal dataset designed to explore the evolution of romantic relationships and family structures within Germany. For this study, we used the most recent release of the Pairfam dataset, which encompasses survey results from 14 waves (Brüderl et al. 2023). Given that the true feelings and thoughts of participants regarding their relationships are unobserved and survey responses are proxies for these sentiments, we treated the evolution of relationships as a confounded Partially Observable Markov Decision Process (POMDP). Our objective was to use the Pairfam dataset to estimate the optimal policy for maximizing long-term satisfaction in romantic relationships.

We defined the immediate reward, R_t, as the mean relationship satisfaction reported by the couple during each survey wave. The continuous action variable, A_t, was selected as the frequency of sharing private feelings and communicating with partners, referred to as intimacy frequency. The observed state variables at each time point, O_t, are represented by a 6-dimensional vector, which includes frequency of conflicts, frequency of appreciation, health status, future orientation of the couple, sexual satisfaction, and satisfaction with friendship of the couple. We defined the reward-proxy, W_t, as a 4-dimensional vector that includes housing condition, household income, division of labor between the couple, and presence of relatives during the interview. Based on romantic relationship literature, the first three variables in W_t could be considered as independent of A_t when conditioned on observed states, and the presence of relatives during interview only becomes available after action is selected. Details on data processing and variable selection can be found in the supplementary materials. To ensure data quality, we excluded samples with more than three missing values for any variable and applied matrix completion methods (Van Buuren and Groothuis-Oudshoorn 2011) to impute the remaining missing data. Consequently, our refined dataset includes 809 trajectories ($n = 809$) each with 14 time points ($T = 14$).

To conduct policy learning on the dataset, we consider the same policy classes introduced in Section 5. We use the same parameterizations, except for the Beta policy class parameterized by a linear basis. Specifically, we define the policy class as Beta $\left(1 + \zeta_1 \operatorname{expit}(\zeta_2^\top o),\ 1 + \zeta_3 \operatorname{expit}(\zeta_4^\top o)\right)$ with actions linearly rescaled to $[-1, 1]$, where $\zeta = [\zeta_1, \zeta_2, \zeta_3, \zeta_4]^\top$, and $o \in \mathbb{R}^7$ is a feature vector that includes an intercept term and the six observed state variables. To ensure numerical stability, we constrain $\zeta_1, \zeta_3 \in (0, 20)$. The forms $1 + \zeta_i \operatorname{expit}(\zeta_j^\top o)$ for $i = 1, 3, j = 2, 4$ are specifically selected for normalization purposes, enabling us to constrain the parameters associated with the beta distribution to be in the range $(1, 21)$, thus, avoiding extremely large or small values that may result in abnormal policies.

We set the discount factor at $\gamma = 0.9$ and employ our proposed method to identify the optimal policy that maximizes long-term relationship satisfaction. To ensure robustness, each simulation randomly selects 400 trajectories as the training data and uses the remaining trajectories for testing data. Specifically, we train the policy on the training data and then apply our proposed policy evaluation method from Algorithm 1 to the testing set to estimate the policy value of learned optimal policy. As shown in Figure 2, our proposed OPE method exhibits the smallest bias in the presence of unmeasured confounders compared to alternative methods.

Figure 3 presents boxplots of the estimated policy values on testing sets for each method based on 50 simulation runs, with the baseline representing the observed discounted return. As

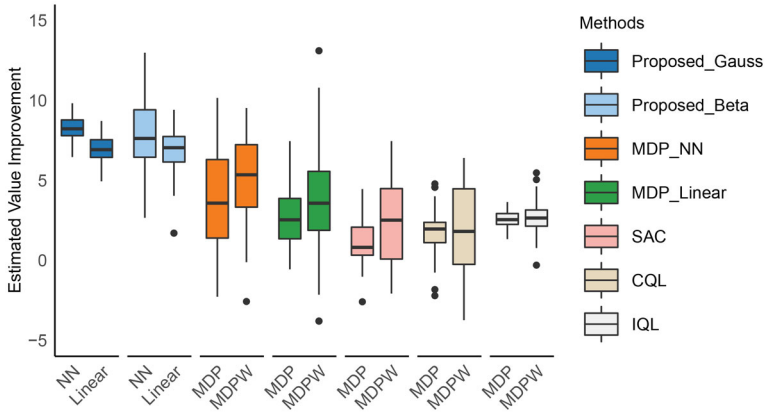

Figure 3. Boxplots of discounted reward improvements under 50 simulation runs. The first four boxes represent the proposed method using Gaussian and Beta policy class with neural network and linear parameterization. The remaining boxes show the MDP and MDPW version of competing methods.

Table 2. Estimated coefficients of the optimal policy.

	Intercept	Conflicts	Appreciation	Health	Commitment	Sex	Friendship
ζ_2	−0.0626	−1.1602	1.4199	1.3774	1.1408	0.9960	−0.2873
ζ_4	−0.8386	1.4781	0.1855	−0.0315	−0.7316	0.3191	−0.6859

Table 3. Observed state variables for each scenario.

Scenarios	Conflicts	Appreciation	Health	Commitment	Sex	Friendship
Scenario 1	0.67	−0.02	0.03	0.51	0.01	0.02
Scenario 2	−0.65	−0.21	−0.12	−0.85	−0.19	−0.14
Scenario 3	−0.78	0.43	0.09	0.52	0.38	0.19
Scenario 4	0.93	−0.66	−0.14	−1.16	−0.61	−0.30

NOTE: All variables have been normalized to have mean 0 and variance 1.

shown in Figure 3, the proposed method achieves the best performance in terms of the improvement on policy value by taking unobserved confounders into account, which is consistent with the results presented in Section 5. In contrast, the MDP-based method tends to overestimate the policy value in model training, which may lead to unreliable policy and is undesirable in safe-critical scenarios. Additionally, we observe that the linear policy class shows similar performance compared to the neural network policy class. Since the linear basis is more interpretable in such a setting, we further examine the learned optimal policy obtained using the linear policy class.

For linear policy class from the Beta distribution, the estimated values of $\zeta = [\zeta_1, \zeta_2, \zeta_3, \zeta_4]^\top$ are shown in Table 2, with $\zeta_1 = 12.3463$ and $\zeta_3 = 7.4619$. To better illustrate the learned policy, we demonstrate the optimal policy distribution under four different observed states. In romantic relationship studies, researchers have classified couples into different categories based on their commitment levels and couple dynamics (e.g., negative interactions) (Beckmeyer and Jamison 2021). Thus, we chose commitment and conflict frequency as a proxy for couple dynamics, and divided all the samples in the dataset into four categories based on whether their commitment and conflict levels are higher than the sample means. As shown in Table 3, the values of observed state variables, other than conflicts and commitment, are selected as the sample means of those variables within each category. It is important to note that all variables have been normalized to have mean 0 and variance 1. Figure 4 shows the induced optimal policy of each observed states.

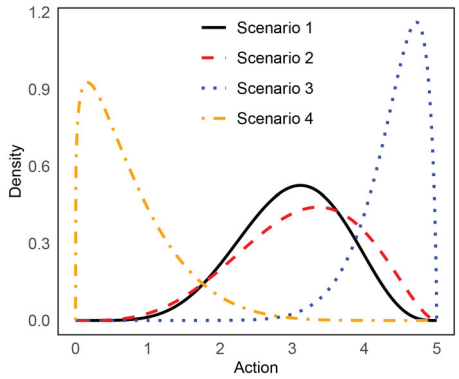

Figure 4. The estimated optimal policy distribution under typical states based on conflict and commitment levels. The corresponding states are defined in Table 3.

Figure 4 suggests that for couples experiencing a high frequency of conflicts and low levels of commitment (Scenario 4), fewer instances of sharing feelings and thoughts should be undertaken to improve satisfaction. In contrast, more frequent sharing is preferred for couples experiencing low conflicts and high commitment (Scenario 3). Additionally, moderate to high levels of sharing feelings are preferred for couples with moderately high commitment and conflict levels (Scenario 1) and couples with moderately low commitment and conflict levels (Scenario 2). These results are consistent with the literature on the effects of sharing feelings and thoughts in romantic relationships. Although relationship researchers generally acknowledge the positive effects of sharing feelings (e.g., openness) (Ogolsky and Bowers 2013), some studies

have found that withholding feelings can be beneficial to relationship quality for individuals with high levels of social anxiety (Kashdan et al. 2007) and for couples with a more communal orientation (Le and Impett 2013).

7. Discussion

In this article, we proposed a novel framework to conduct offline policy evaluation and policy learning on continuous action space with the existence of unmeasured confounder. We extended the proximal causal inference framework (Cui and Tchetgen Tchetgen 2021) to an infinite horizon setting for the identification of policy value for a fixed target policy, and developed the corresponding minimax estimator. Based on the proposed estimator, we further developed a policy-gradient based method to search for the in-class optimal policy. The PAC bound of the proposed algorithm is provided to analyze its sample complexity.

Several improvements and extensions are worth exploring in the future. First, we assumed there is no coverage issue for the batch dataset. However, this assumption may not hold in real-world scenarios, such as medical applications where sample sizes are generally small and data is expensive to obtain. Thus, properly addressing the data coverage issue under the POMDP setting for both policy evaluation (Zhang and Jiang 2024) and policy learning would be an interesting topic. Second, as demonstrated in real-data application, different observed state variables play various roles in formulating the policy. Meanwhile, conducting variable selection in an offline RL setting remains a challenging task, as there is no ground truth available for performance comparison. Systematically studying this problem would greatly improve the generalization of RL techniques. The proposed method can also be extended to multidimensional action spaces. A common approach is to approximate Gaussian policy parameters using neural networks (Haarnoja et al. 2018; Feng, Li, and Liu 2019). Finally, the proposed algorithm requires relatively large computation and memory resources due to the necessity of evaluating a new policy at each iteration. Therefore, developing a more efficient algorithm is desirable. A possible approach is to directly identify the policy gradient under the POMDP (Hong, Qi, and Xu 2023), however, such identification in continuous action space remains a challenge. Another computational solution is Yang, Pilanci, and Wainwright (2017) which can improve the time and space complexity in such RKHS based approaches.

Supplementary Materials

The supplementary material contains the technical proofs of all main theorems, along with additional details and results from numerical experiments.

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Generative AI tools were used to edit grammar and enhance presentation.

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The authors report there are no competing interests to declare.

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